

Teams: Jason Lee(40346888)

Haolei Ma(40316304)

For Project 2, we decided not to start from scratch but instead build on what we had already created in Project 1. The core idea stayed the same, it was a simple incremental clicking game inspired by games like *Cookie Clicker*, where the core interaction revolves around repeating an action and gradually accumulating results. But we focused much more on making the experience feel interactive and engaging for users. To achieve that, we used external libraries like anime.js, particles.js, and three.js, which helped us create more dynamic visuals and a smoother, more immersive experience. This direction connects with ideas from *Interaction Design: Beyond Human-Computer Interaction*, which highlights how important continuous feedback is for keeping users involved.

We mainly used anime.js to handle animations. For example, the randomly appearing images now scale, rotate, and fade in, making them feel more lively and less static. We also replaced the original CSS animation for the error message with anime.js, which gave us greater control over timing, movement, and overall behavior.

Particles.js allowed us to create a more interactive background. The particles respond to user actions like clicks and cursor movement, which makes the environment feel more responsive. We also experimented with changing particle behavior over time. Once the user reaches a certain number of clicks, the system enters a kind of “collapse mode,” where everything speeds up, becomes more chaotic, and looks more intense, almost like the system is breaking down.

We also added more layers of feedback tied directly to user interaction. Each click triggers visual responses such as particle bursts, background color changes, and occasional background images. Every five clicks, a random image appears, adding a sense of unpredictability. As users keep clicking, the effects become more dramatic, including things like screen shaking and warning messages, which build a stronger sense of escalation. This idea of increasing intensity through repeated interaction is similar to feedback loops discussed in *The Art of Game Design: A Book of Lenses*, where ongoing rewards motivate users to continue engaging.

We chose Three.js as the focus of our core update. We added a cube to the 3D container to serve as our new button, replacing the previous flat button, and transferred all interactive functions to it. Building on this, we've also added some visual and interactive feedback to the cube itself: In any mode, the cube will rotate on its own at regular intervals without any user interaction; when a click occurs, the cube will shake in sync with the speed and rhythm of the click. When the number of clicks reaches 30, the cube will explode and shatter as the system warning appears, but it will leave a cube remain in the same location so that the user can continue to click on it. When the number of clicks reaches 50, the cube will refresh along with the system.

When we did our presentation on Project 1 a few weeks ago, we mentioned that we wanted to adapt it for mobile devices. We achieved this goal, at least “superficially”, by using JSON and Server Worker Registration. The reason to say this is that we don't actually have a true app backend; we simply use server registration to cast the desktop web page onto the mobile device, thereby simulating the mobile app experience as closely as possible. The app logo that users see on the home screen is a PNG file we imported in advance. We have also implemented corresponding functional changes to accommodate the transition from mouse clicks to finger

taps. If any users would like to try out our mobile apps, step by step instructions are provided at the end of this statement.

Overall, these improvements were meant to make the project feel more alive and reactive. Compared to Project 1, this version is much more engaging because it constantly responds to user input with increasingly noticeable and complex effects.

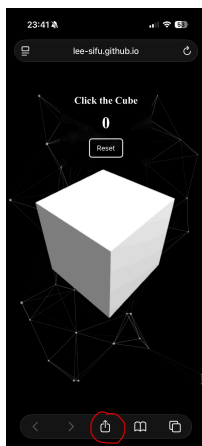
Mobile App Instructions

P.S: Since all our team members use iPhones, this instruction currently only supports app generation on iOS. We sincerely apologize to users who use Android phones but wish to experience the mobile version. 🙏

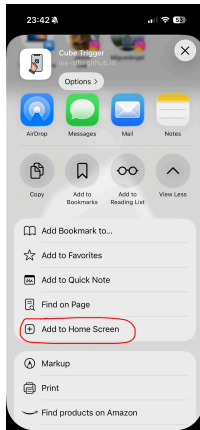
1. Open GitHub web page link with Safari:

https://lee-sifu.github.io/Project-2_Cart263/Project-2_Cart263-main/

2. Find share button at the bottom



3. Find “Add to Home Screen”



4. The icon of the app will show on your home screen. Enjoy!

References

https://digilib.stiestekom.ac.id/assets/dokumen/ebook/feb_d2da2b2ae5541cebf7e87884e0a46b395eaff87a_1659872033.pdf

Helen Sharp, Yvonne Rogers, Jennifer Preece, Interaction Design: Beyond Human-Computer Interaction, John Wiley & Sons, 2023

https://rsd2-alert-durden-connections.weebly.com/uploads/6/7/1/6/6716949/3-the-art-of-game-design-a-book-of-lenses_3.pdf

Jesse Schell, The Art of Game Design: A Book of Lenses, CRC Press, 2019

Screen Grabs

